

For the balanced reaction above, how many grams of nitrogen monoxide are produced when 16.0 g of oxygen react?

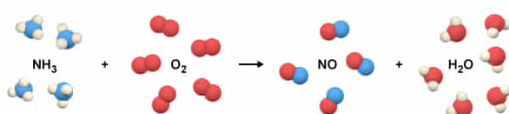
- ✓ ☒ A. 12.0 g [57%]
☐ B. 13.6 g [12%]
☐ C. 18.8 g [8%]
☐ D. 24.0 g [20%]

Correct

57% answered correctly

Explanation:

Formation of nitrogen monoxide from ammonia



Mass O ₂ reacting	Molar mass of O ₂	Molar ratio of reaction	Molar mass of NO	Mass NO produced
16.0 g O ₂	$\frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2}$	$\frac{4 \text{ mol NO}}{5 \text{ mol O}_2}$	$\frac{30.01 \text{ g NO}}{1 \text{ mol NO}}$	12.0 g NO

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The **molar mass** of a compound is the amount of **mass contained in 1 mole** of the compound (grams per mole) and can be used to convert the mass of a compound to the number of moles. The reactants and/or products of a particular reaction are related to one another by the **mole ratios** (**stoichiometry**) from a **balanced reaction equation** or chemical formula.

For the reaction in this question, if 16.0 g of O₂ react, the number of moles of O₂ participating in the reaction can be calculated by dividing the mass of O₂ by its molar mass to give 0.500 mol O₂.

The balanced reaction equation shows that 4 moles of NO are produced for every 5 moles of O₂ that react. When this stoichiometric ratio is applied, the number of moles of NO produced from 0.500 mol O₂ can be found: 0.400 mol NO.

The mass of the 0.400 mol of NO produced in the reaction can be determined by multiplying the moles by the molar mass of NO (30.01 g/mol). Therefore, 12.0 g of nitrogen monoxide are produced when 16.0 g of oxygen react.

(Choice B) 13.6 g is obtained if the numerical values in molar mass ratios are inverted (as in 32.00 mol/g and 30.01 mol/g) during the first and last steps of the

calculation.

(Choice C) 18.8 g is obtained if the stoichiometric mole ratio is inverted during the second step of the calculation.

(Choice D) 24.0 g is obtained if 16.00 g/mol (the molar mass of an oxygen atom), is used instead of 32.00 g/mol (the molar mass of an oxygen molecule) during the first step of the calculation.

Educational objective:

The molar mass (the amount of mass in 1 mole of a compound) permits conversions between a given amount of mass and the number of moles that it contains.

Stoichiometric mole ratios from a balanced reaction equation can be applied to determine the number of moles of one chemical species relative to the moles of another chemical species in a reaction.

Time spent:0 sec

Subject:
General Chemistry

QID:402735

System:
Interactions of Chemical Substances